Pasture and Fodder Production and Utilisation Manual for Livestock Farmers in Zambia









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1.0 Pastures and Fodder Crops for Livestock Production

Livestock production in Zambia is very important. The common type of grazing animals(ruminants) reared are cattle, sheep, goats and donkeys. For good productivity of the grazing animals they need a very good source of grazing materials. Grazing materials consists of grasses, legume, trees and browse species.

Value of Pastures, Fodder Crops and Browse trees

a. Feed for animals.
   Pastures and fodder crops constitute the major source of nutrients in the diet of cattle and goats. The digestive system in these species is evolved to effectively use these materials, and actually has a requirement for a minimum of these materials in their diets.

For animal grazing



For Browse



b. Other uses in animal production
   i) Used for proving shade for animals especially during the dry and hot months of the year.
   ii) Medicinal use. Some plants such as neem tree and tephrosia can be used in controlling external parasites such as ticks.
   iii) A number of pasture species are used to add beauty to the areas around homesteads and surroundings.

For shade



Not all trees should be cut in grazing areas. Some should be left for shade.

c. Soil protection and improvement
Improve soil fertility by fixation of nitrogen by legumes, by provision of organic matter and humus. Organic matter or Humus is:

(i.) a source nutrients for plants and for micro-organisms,
(ii.) helps reduce surface run off of water,
(iii.) as cover crops prevents soil erosion
(iv.) it increases the infiltration rate,
(v.) it improves the water holding capacity,
(vi.) it improves soil structure and enhances good species.

Soil cover

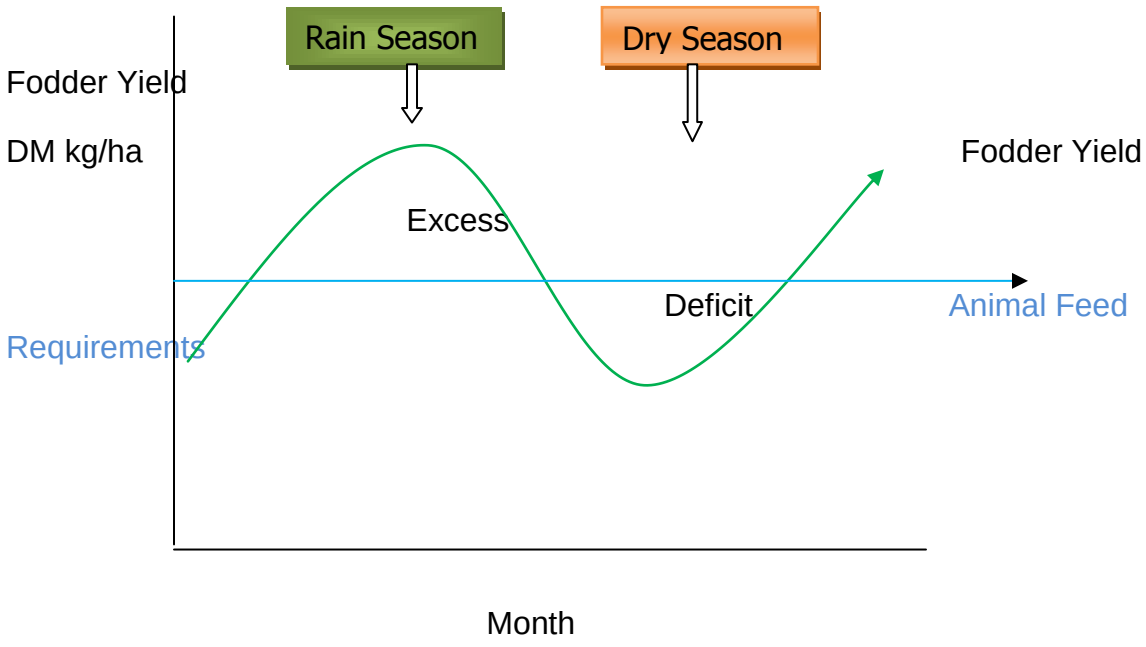


- ✓ Soil cover to prevent soil erosion
- ✓ They also act as carbon sinks which help reduce the amount of green house gases in the atmospheres thereby reducing global warming

1.1 Rainfall and Pasture Availability

The rainfall in Zambia is mono-modal (one rain season in a year). During the wet (rain) season these lands provide more than adequate forage to maintain productive animals. In the dry season however, the quantity and quality of forage greatly decreases and is generally low in nutritional value (Figure 1). Livestock sustained on such diets often lose weight and productivity and in times of drought many animals die.

Figure 1 Pattern of Pasture availability in Zambia



To ensure that there is a continuous supply of feed throughout the year the excess feed during periods of excess should be conserved for feeding during the dry season. By conserving and storing some of the feeds available during periods of surplus, livestock keepers can ensure that they have sufficient feed to offer their livestock throughout the year.

1.3 Types of pasture and fodder crops

The main pasture and fodder crops comprises grasses, legumes and browse trees and shrubs



Grasses

- ✓ Differ widely in size shape and habit of growth
- ✓ May be annual or perennial.
- ✓ They are rich in carbohydrates
- ✓ Examples : Rhodes grass, star grass, Buffel grass, Elephant (Napier) grass, Guinea grass, Cow candy and many more



Legumes

- ✓ Different types
- ✓ All bear pods
- ✓ Are able fix nitrogen from the air into the soil through root nodules.
- ✓ Legumes are high in protein and minerals
- ✓ Example: Silver leaf Desmodium, Leucaena, Siratro, Lucerne and Stylo



Fodder trees and shrubs

- ✓ browsable plants that are palatable to animals.
- ✓ Leguminous and non-leguminous types
- ✓ have palatable leaves, flowers, small stems and twigs
- ✓ remain green in the dry season and have higher nutrition than grasses

Other grazing materials for ruminants in clued forbs, sedges and rushed



Forbs

- ✓ Forbs are herbaceous (not woody), broadleaf plants that are not grass-like.
- ✓ An example of a common forage forb in Zambia is sunflower plant



Sedges

- ✓ superficially resemble grasses but the difference is that their stems have triangular cross-sections (with occasional exceptions)
- ✓ Sedges are perennial plants that are commonly found in shallow water or moist soils They often grow in thick cluster
- ✓ They are found along any rivers and flood plains in Zambia.
- ✓ Often used as dry season grazing of animals



Rushes

- ✓ They are slow-growing, rhizomatous, herbaceous plants,
- ✓ they may superficially resemble grasses and sedges.
- ✓ In Zambia they grow mostly in wetland habitats.
- ✓ A few rushes are annuals, but most are perennials.
- ✓ They are important as feed for cattle in the dry seasson

1.4 Establishment of Improved Pasture and Fodder Crops

The purpose of growing forages is to provide good and palatable succulent feed for livestock. Forages have establishment problems which are unique and not common to field crops. For example they have very small seeds and in growing/planting, precision is very important; They may take one year or more to establish; Once seeded and stand established, they are used for years giving problems in maintaining the stand e.g weed problems; and Very often they are established in impure stands thus weed and disease control become difficult and complex

What steps do farmers need to follow in order for them to establish improved pasture and fodder crops on their farms?

Just like when growing food and cash crops, you need to follow at least three or four basic steps for establishment of improved pasture and fodder crops. The first step is to start with land clearing and soil cultivation for the preparation of a fine seed bed. Depending on whether one is planting improved pastures from seeds or vegetative planting materials, the soil must be prepared accordingly. Seeds need a finer tilth whereas vegetative planting materials can tolerate a coarser land preparation.

Note that land preparation can be done manually or mechanically depending on the area being cultivated. Which ever the method, you should start with clearing the area of all debris and stumps followed by cultivation to break the sod. All big clods and stones must be removed to obtain a fine tilth.

Once the land is prepared, you must ensure the soil has got adequate moisture content before planting to maximize on the chances of germination or propagation. This means either proper timing of the rain before planting or where possible irrigating the prepared portion of land. Temperature is usually not a factor in the germination of pasture seeds in Zambia and many other tropical countries.

1.4.1 Land preparation

Plough early to kill off weeds

Hand hoe preparation



- Cheaper method but takes a lot of time to cover a reasonable area
- Ideal for small scale fodder production or where it is cheaper to hire labour

Ploughing using ox drawn plough



Ideal where there are a lot of draught animals

Cover s more area within a given period of time compared to using hoes

.

Ploughing with tractor



- Quick method and covers a large area within a short time
- Requires a lot on financial input
- Ideal for large scale pasture production



- ✓ Disc break down large clods.
- ✓ Big clods reduces germination of seeds especially small ones



- ✓ Need a well-prepared seedbed.
- ✓ Small seeds will need a finer seedbed
- ✓ Roll to give a firm seedbed

1.4.2 Planting methods

Drilling

- ✓ more perfect method for pasture establishment than broadcasting
- ✓ the seeds can be placed at a desired depth and in closer proximity to the fertilizer.
- ✓ Small-seeded grasses and legumes should be sown at depths not exceeding 1cm.
- ✓ Large-seeded legumes such as leucaena can be sown deeper up to 5cm deep.

Holes can be made in rows



1,2 or 3 seeds can be placed in ech hole





- ✓ Making lines can be done using ox or tractor drawn implements.
- ✓ Seed can be planted at the desired depth and spacing



- ✓ Row planting makes it easier to weed
- ✓ Seeds are uniformly spread over the whole field
- ✓ As the plants grow they can sometimes cover the area between the rows and they appear as though they were not planted in rows



Pigeon peas planted in rows



Rhodes grass planted in rows



Napier grass being planted in rows.

Broadcasting



- Broadcasting by hand or fertilizer spreader is the easiest method of sowing pasture seed.
- A more distribution of the seed is obtained by mixing it with sand, sawdust or fertilizer.
- If the seed has been broadcast onto a prepared seedbed, it should be covered lightly by harrowing or by rolling a tree branch over the soil.



- Star grass planted by broadcasting

1.4.3 Planting materials

A variety of planting materials can be used to establish pastures. They include use of seeds and vegetative propagation using different other plant parts.

Seeds types

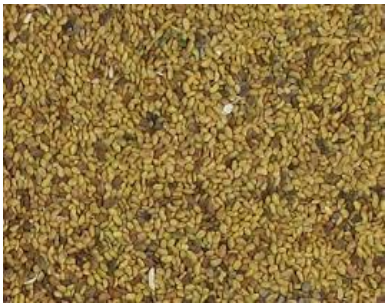
Small seeds



Rhodes grass



Sunhemp



Lucerne

Small-seeded grasses and legumes should be sown at depths not exceeding 1cm.

Large seeds



Velvet Beans



Jack Beans



Dolichos Lab lab

Large-seeded legumes such as leucaena can be sown deeper up to 5cm deep

Vegetative Planting materials

Root Splits



Root Splits - this is where the whole plant is dug up and the crown area is split up (Figure 3.1). Then the root splits are planted. This method can be used in grasses that do not produce seed



Napier grass etc

Stem Cuttings

Star grass



Stem cuttings - Stem cuttings are the most commonly used method of vegetative propagation. A stem cutting is plant stem including a tip (e.g. shoot, twig, sucker,) or a portion of a stem without the apex that includes one or more nodes removed from a parent plant and capable of rooting

Stem Cuttings

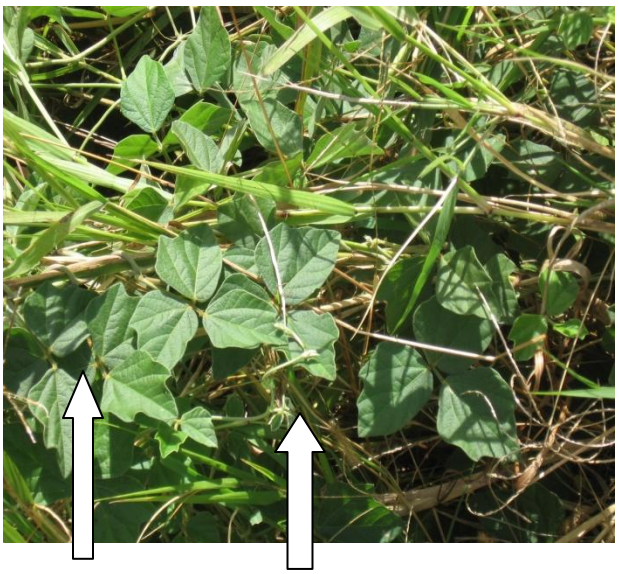


Napier grass stem cuttings

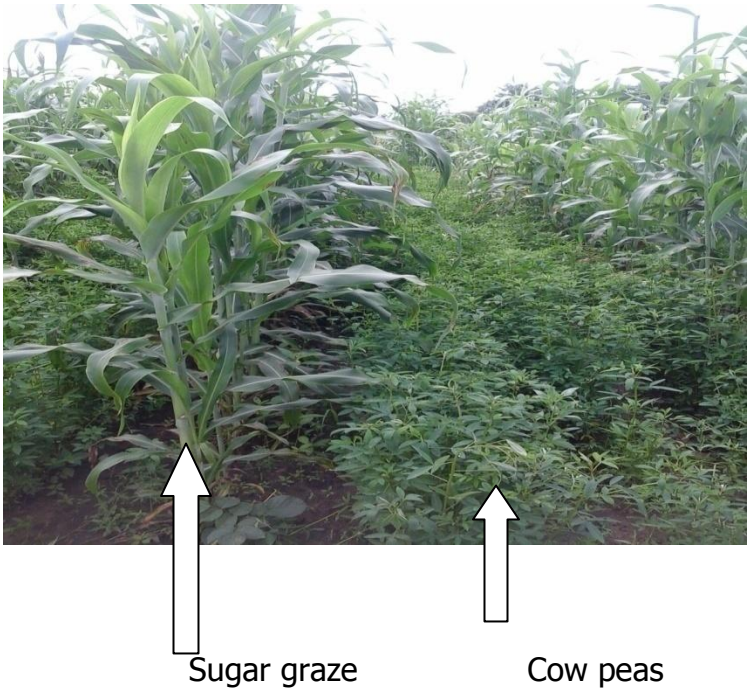
1.5 Growing mixtures of grasses and legumes

Species and Cultivar Selection

It is usually desirable to seed land intended for permanent pasture with a single species because establishment and management is easier. However, the usual thing is growing them in impure mixtures (grasses and legumes mixtures).



Siratro Star grass



Sugar graze Cow peas

Advantages of Mixtures

- ✓ Legumes can fix adequate nitrogen
- ✓ Legumes have high protein
- ✓ Mixtures have high production
- ✓ mixtures biologically control weeds.
- ✓ Mixtures provide a better balanced ration since legumes are richer in both proteins and minerals.
- ✓ Mixtures are less likely to cause bloat than pure legumes.

Guidelines

- ✓ Keep mixtures simple (one grass one legume
- ✓ They must have similar maturation dates



Maize and Pasture legume mixture



Sugar graze and cow peas mixture



Napier grass and Siratro mixture

Mixed Grass-legume Pasture

Where mixed grass-legume pasture is to be established, two methods of seeding are available;

a. sow the legume three to four weeks before the grass; this will allow the slower growing legume to establish before the grass.
b. Sow the grass and legume in alternate pure strips two to three meters wide.
c. different pasture seeds (grass and legume) can also be blended in the same bag before planting

1.6 Seed Treatment

1. Scralification to break dormancy.

a. Mechanical - this can be done by hand rubbing with sand paper or pricking the seed coat with a pin or a sharp pointed knife. With small seeds this may be arduous and time consuming. Small and large scale scarifying equipment is available.



b. **Acid Treatment** - Placing seeds in concentrated sulphuric acid for 3-5 minutes followed by thorough washing and drying is an effective means of scarifying. Extreme caution is required so as not to splash the acid on the body and clothing.

c. **Hot water treatment** - Soaking seeds (e.g *leucaena*) in water heated to about 80ºC for 5 minutes, rapid cooling and drying improves germination.

2. Inoculation

For some legumes seed inoculation with the appropriate rhizobium ensures that bacteria necessary for nodulation is introduced at the time of sowing. Herbage legumes grown in many parts of the tropics however produce nodules without inoculation because the cowpea-type rhizobia is naturally present where native or naturalized legumes are growing or were recently grown. Inoculated seed should not be dry planted.

3. Pelleting

Seed pelleting is the coating of grasses and legumes seeds with a layer of fine powder held on to the seed by a sticker. This is done to ensure even flow of seed from the drill or aerial applicator. This also aids in inoculation of the seed.

4. Seed Dressing

Seeds are dressed with fungicides and pesticides so as to protect the seeds from various fungal diseases and damage by insects and ants when the seeds are in storage or are planted in the field during early seed establishment. This also offers an opportunity to include other chemicals that can enhance the growth of these pastures like some micronutrients.

1.7 Seed Rates

The actual rate used may vary depending on a number of factors;

a. Rates for drilled seeds are usually lower then those for broadcast seeds.
b. If grass is to be sown with a legume, the seed rate for the grass should be reduced slightly.
c. If the germination percentage of the seed is low, the seed rate should be increased.
d. When a mixture of legumes is sown, the rate of each legume can be reduced so that the total rate is about 4-6 kg/ha.
e. Seed rates for legumes with large seeds could be as high as 20 kg per hectare

1.8 Manuring and Fertilizing

Pastures can be strategically and profitably managed by well-planned fertilization and production management programs. Some plant nutrients are required in very small amounts and are adequately provided from the soil. However, others, such as nitrogen (N), phosphorus (P), and potassium (K), are required in greater amounts and are commonly applied as fertilizer amendments. Soil sampling and testing are used to determine the availability of P and K nutrients, and the amount that might need to be added to make up for any deficiencies. If insufficient amounts are available in the soil or supplied with fertilizers or manure, both pasture and animal production may be restricted, and profitability may be limited. Rather than just depending on chemical fertilizers, one should always consider use of animal and compost manures. If you have manure (kraal, chicken, sheep or from goats) make the best use of it – do NOT let it dry out, but keep it covered or bury it in basins or rip lines as soon as possible. If you have the time and materials make compost but find out how to do it effectively.

Fertilizing Grass Pastures

How much nitrogen to apply For grass pastures to be productive, first priority should be given to meeting N needs. Grasses respond quickly to N when other growing conditions are good. However, N fertilization is not as profitable if P and K supplies are inadequate. suggested recommendations are that one needs to apply 60kg N, 30kg P and 30kg K for establishing a hectare of grass forages (Nagib and Eng, 2012).

Fertilizing Mixed Legume-Grass Pastures

Nitrogen for mixed legume-grass pastures Nodulated legumes can capture N gas from the air and symbiotically "fix" it in plant-available forms for their own use and for grass growing in mixtures with the legume. If the legume portion is less than one-third, the grass in the mixture is likely to respond significantly to added N fertilizer at rates described for grass pastures. If the forage stand is one-third or more legume, no additional N fertilizer application is generally recommended.

Phosphorus and potassium for legume-grass pasture Legume or legume-grass pastures have a higher requirement for P and K than do grass pastures. Adequate P and K are essentials for legume establishment, stand maintenance, and satisfactory yields. These two nutrients maintain legume yields, and also enhance disease resistance, winter hardiness, and longer stand life. For grass legume mixtures, the recommendation is that you use 30kg of Potassium and 30kg Phosphorus

Fertilizer will stimulate weed growth as well as grasses and legumes in a pasture. Clip to control weeds or spray grass pastures with labeled herbicides. Spring spraying is the most effective time to control susceptible annual broadleaf weeds.

1.9 General Management Practices

Once the pasture and fodder crops have been established, the next step is management that includes weeding, pests and disease control and fertilization to see the crop to maturity!



Weeding Velvet beans



A clean field of young Rhodes grass

2.0 Important Pasture grasses

2.1 Rhodes grass (*Chloris gayana*)



Rhodes grass is a tufted stoloniferous leafy perennial plant 0.3 to 1.2 m or more in height.

Mostly grown for direct grazing or hay making



Seeds have extending appendages called awns.

Three strains of Rhodes grass namely; Katombora Rhodes, Boma and Mbarara Rhodes. Boma and Mbarara are giant types

Propagation

a. Grows very well in loamy soils.
b. Can withstand flooding up to 15 days.
c. Relatively drought resistant can grow at annual rainfall of 600mm
d. Tolerant to fire.

Land Preparation

- Seedbed should be with fine tilth.



Seed Rate.

a. Rows or broadcast seed rate can be 5-10 kg per ha.
b. Drilled seeds no more than 1 cm deep.
c. Sow rows 0.5m to 1.0m part.
d. Sowing should not take place until the rains are well established.
e. Germination improved if seed is stored for at least 6 months before planting.

If germination is poor the stand should be allowed to flower and set seed before the first grazing is taken



Rhodes grass planted in rows

Fertiliser Requirements

a. Establishment year
   - 400-600 kg single super phosphate per ha
   - and 150 kg ammonium nitrate per ha.

b. Annual maintenance
   - 200-300 kg single super phosphate per ha
   - and 150 kg ammonium nitrate per ha
   - broadcast at start of rains each subsequent year.

c. Apply nitrogen at about one month intervals after commencement of grazing
d. Rates of 120-170 kg ammonium nitrate/ha/month.
e. Urea not recommended as field losses of nitrogen can be high.
f. Liming of the soil at a rate of 1-2 tons/ha may be beneficial if the soil ph falls below 4.6

Management

a. Rhodes grass seedlings are very small
b. Weed competition is usually the main difficulty at establishment
c. cutting or grazing can reduce this competition.
d. No known specific diseases

Yield

It yields between 12-24 tons of dry matter per ha in well fertilized soils.

Utilisation

1. For grazing pasture.

    a. It can withstand heavy grazing therefore can be heavily stocked
    b. Should not be grazed below 15-20 cm from the ground
    c. Rotational grazing is recommended at least three weeks resting period between grazing.

2. More suitable for hay making than grazing due to lack of persistence on trampling. Cultivars Boma and Mbarara good for hay and silage



Square bales



Round Bales

3. Grown in mixtures with legumes.
   -Mixtures with legumes give higher animal production than pure Rhodes grass

4. Suited for short tem pastures on a cropping rotation since it is easy to eradicate.
5. In Zambia Katombora cultivar is resistant to root knot nematode and is therefore a suitable grass for growing in rotation with tobacco.
6. Sale of seed can bring very good revenue

2.2 Star Grass (*Cynodon plectostachyus, nlemfluencis*).



a. Is a tufted perennial
b. Is spread by stolons which are long internodes from which roots sprout
c. Is adapted to warm regions with little or no frost.
d. It does well on granite sands
e. Is drought resistant.
f. Very palatable and nutritious.
g. Responds better to irrigation.
h. very competitive grass and persistent (very difficult to eradicate).

Propagation

a. Plough,
b. Prepare seedbed.
c. Can be established from rooted runners or from division of root stock.
d. The best results are obtained from hand planting at a spacing of 0.5m x 0.5m.
e. This method is labour intensive

Alternative planting methods

a. Scatter runners at a high density over the area and disk in the soil along with the fertilizer. A light rolling will be beneficial.
b. Drop runners into a plough furrow, leaving some of the material protruding and cover over.
c. For best results plant using runners from the current season`s growth.
d. Plant when rains are well established.
e. Light graze when vigorous growth shows to encourage lateral spread of runners.
f. After establishment is complete it can be heavily grazed and this is essential for good production.

Fertiliser Requirements

a. Apply nitrogen at about one month intervals after commencement of grazing
b. Rates of 120-170 kg ammonium nitrate/ha/month.
c. Urea not recommended as field losses of nitrogen can be high.
d. Liming of the soil at a rate of 1-2 tons/ha may be beneficial if the soil ph falls below 4.6

Yield

a. Dry matter yield of up to 12 tones/ha in the rain season and
b. About 8 tones/ha in the dry season

Utilisation

a. Grazing. Very good as a permanent pasture. Can maintain seven or more steers/ha over the main growing period with average gains of more than 700grams/steer/day.
b. Very good for hay, green chop and silage.



Star grass grazing

2.3 Guinea Grass (*Panicum maximum*)





a. a tall tufted perennial of 0.5-4.5m high (extremely variable in height).
b. Very variable specie and about a dozen botanical varieties have been named.
c. Two distinct untypical forms:
    i. Green Panic or Petrie and
    ii. Embu found in Kenya
d. Grows in warm frost free tropical climate.
e. Root system is deep - some tolerance to drought.
f. Adapted to wide range of soils.
g. With-stand burning
h. Extremely tolerant to shade
i. Cannot survive prolonged water logging
j. Responds well to irrigation during the hot and dry months (September to November)

Propagation

a. Land preparation
b. Plough disc and make proper seedbed.
c. Seed Rate: at a seed rate of 3-7 kg/ha (as for Rhodes grass) fresh seed should not be sown. Plant just before the rain season.

Fertilizer.

a. Establishment year

    i. 400-600 kg single super phosphate per ha
    ii. and 150 kg ammonium nitrate per ha.

b. Annual maintenance
    i. 200-300 kg single super phosphate per ha
    ii. and 150 kg ammonium nitrate per ha
    iii. broadcast at start of rains each subsequent year.
c. Apply nitrogen at about one month intervals after commencement of grazing
d. Rates of 120-170 kg ammonium nitrate/ha/month.
e. Urea not recommended as field losses of nitrogen can be high.
f. Liming of the soil at a rate of 1-2 tons/ha may be beneficial if the soil ph falls below 4.6

Yield

- Yield 12 tonnes of dry Matter per ha.

1. For grazing. Very palatable. Rotational grazing is recommended with at least three weeks resting period between grazing.
2. Cures well as hay or silage and Has very high CP 10 – 20%
3. Combines well with legume mixtures.
4. It responds very well to nitrogen fertilizer than a number of other tropical grasses.

2.4 Napier Grass (*Pennisetum purpureum*)



Napier grass is one of the popular crops grown for Zero grazing on smallholder dairy farms!

a. A tall deep rooted grass similar in appearance to sugar cane.
b. one of the highest yielding grasses and does well under irrigation.
c. Can grow up to 2 meters in height
d. Is generally cut for silage or green feed but can be grazed.
e. Is drought resistant and moderately frost resistant.

Propagation

a. from stem cuttings or root splits.
b. Old woody stems do not make good planting material.
c. Root splits give better results.
d. Stem cuttings should contain three to four nodes**.**

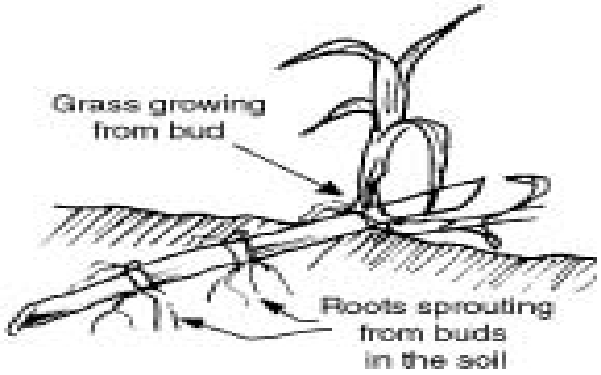

Stem Cuttings from Napier grass

Planting

a. Cuttings should be placed at an angle in the ground such that two or three nodes are buried in the soil and one or two remain uncovered.

b. Plant in rows 0.5m apart and 0.3-0.9 within rows

c. Replanting should be done after 3-4 years







Fertiliser Requirements

- In the establishment year 500kg/ha D compound .

- Annual maintenance of 200-300kg/ha D compound at the start of the rains and 120 kg/ha ammonium nitrate after each harvest

Utilisation

1. Fed to animals as green chop.
    a. Cut when 1m high. Beyond this unpalatable.
    b. Cut to the height of 20 from the ground. Below this, re-growth reduced and delayed.
    c. cut grass and chop in small lengths and given to the animals.
    d. Grazing is not recommended. Old cut stems and sharp midrib of broken leaves injure animals.





2. Napier grass also makes excellent silage..

3.0 Important Pasture Legumes

3.1 Alfalfa or Lucerne (*Medicago sativa*)





a. Is a herbaceous perennial legume.
b. Mature plants may have 5-25 stems
c. Grows to a height of 60-90cm.

Propagation

a. Have a fine as the seed is small.
b. Shallow planting on heavy soils
c. Grows on a wide range of soil type, from sandy to clay soils.
d. Requires good drainage.
e. pH not lower than 5.0 (acidic).
f. Optimum periods for sowing are between March and June.
g. High temperatures and high humidity, diseases and weed competition can severely effect establishment and growth of young Lucerne if the crop is sown during the rain season.
h. Recommended sowing rates are between 15-25kg per ha.
i. Seed can be drilled in rows or broadcast
j. Cover seed no more than 2cm

Inoculation

- strongly advisable to artificially inoculate seed prior to sowing

Fertiliser Requirement

Application Kg Per Ha

a. Light Textured Soil: Basal 80 -100 kg A or V or D 30 solubor
   Annual Maintanance: 300 – 400 kg A or V or D

b. Heavy textured soil: Basal 600-800 kg A or V or D. 200 Solubor
   Annual Maintanance 200 – 300 kg A or V or D

It has nutrient deficiencies of Nitrogen, Phosphorus, Boron and Sulphur

Weed control

a. Weed can be a major problem particularly during the rains
b. The presence of weeds will reduce Lucerne growth and lower quality.
c. Toxic weeds may poison stock.

Pests and diseases

a. Has few pests of importance in Zambia.
b. Diseases of importance only during the rains.
c. Common diseases are common leaf spot, anthracnose, rust.
d. Frequent cutting is the only practical means of control.

Yield

a. Highest a yields and quality are obtained during the months May to October.
b. Yields at early flower of 2.0 to 4.0 tonnes dry matter per hectare
c. Per cut (12-20 tonnes feed) are possible, depending on time of the year and age.
d. Can persist for 3-4 years before replanting.

Alfalfa is one of the highest yielding forage legume (20 t/ha DM in the USA, about 16 t/ha DM in France). Under irrigation, it can produce 25 to 27 t/ha DM with a production reduced in the 3rd year to 8-15 t/ha DM

Utilisation

1. It is a high value fodder crop because of the following reasons
   a. It has high Crude protein value lies between 18-25 percent
   b. Intake and digestibility better than for grasses
   c. Is high in all essential minerals except for phosphorous and possibly iodine.
   d. It is a rich source of carotene and most vitamins.

2. Pasture –not recommended to allow animals to graze because of danger of bloat.
3. Green Chop – this is the best way to feed animals
4. Silage making – the process of silage making lowers feeding value
4. Hay – the process of hay making lower feeding value



Lucerne chopped and given to dairy cattle

3.2 Siratro (*Macroptilium atropurpureum*)





a. Perennial herb with deep, swollen taproot and trailing, climbing and twining stems

b. Siratro tolerates dry conditions because of its deep taproot and its ability to reduce evapotranspiration, either by virtue of pubescent leaves or by shedding its leaves
c. Mainly used for permanent and short-term pastures.

Propagation

a. grows on a wide range of soils, from deep sandy to light clayey soils, provided they are well-drained.
b. It tolerates a wide range of soil pH (4.5-8)
c. one of the most tolerant tropical legumes to Al, Mn and NaCl.
d. Siratro is a N-fixing legume and does not require added N fertilizer.
e. It responds positively to additions of P and K fertilizer in sandy soils so basal application of 200kg/ha of single super phosphate would be good
f.
g. It responds well to warm conditions.
h. Also can survive frost and fire
i. Siratro is nearly always grown in grass mixtures.
j. Fresh seed should be scrarified but inoculation is not required.
k. Seed rates 3-5kg /ha.
l. It can naturally regrow from seeds after fire.
m. Seedbed preparation and seed scarification can facilitate establishment.
n. Over-sowing siratro in existing pastures is possible.
o. It has been successfully grown with Rhodes grass, Buffel grass, green panic, *Setaria sphacelata*

Yield

a. Herbage yield : 5-10 tonnes/ha DM
b. remains productive for 5 to 7 years.

Pasture

Siratro is sensitive to overgrazing,
it is thus recommended to adjust stocking rates or to use a light grazing system to prevent its rapid disappearance.
Once established, siratro can be moderately or rotationally grazed.
A grazing height of 10-15 cm is best to encourage regrowth while lower cutting heights hinder fodder yields.
A good rotational grazing management would combine a 2-week grazing period and a 4 to 6-week rest period.

3.3 Silverleaf Desmodium (*Desmodium uncinatum*)





Desmodium uncinatum is a warm season legume

a trailing perennial legume that may grow up to several meters long over surrounding vegetation.

It has a large but shallow root system.



The leaflets are ovate, 3-6 cm long, 1.5-3 cm broad, dark green with a silvery midrib at the upper side.

Propagation

a. Optimal growth is obtained at average temperatures ranging between 25 and 30°C.
b. Annual rainfall should be about 1000 mm.
c. *Desmodium uncinatum* grows on a wide range of soils from sands to clay loams, but does better on light, friable soils.
d. It can withstand moderate acidity (down to pH 5) and soils with high Al and Mn contents.
e. It is somewhat tolerant to flooding and drought (up to 3 months) but cannot stand poorly drained soils, such as clays, or soil salinity.

Sowing

a. Silverleaf desmodium can be sown in pure stands or in mixed pasture with grasses or other legumes.

b. It is suitable for intercropping with cereals and fodder crops that provide shade and support to its vines.
c. It does well with other legumes such as greenleaf desmodium (*Desmodium intortum*) and siratro (*Macroptilium atropurpureum)*, and with cold tolerant grasses such as the giant setaria (Setaria sphacelata var. splendida).
d. It can be grown in association with tropical grasses such as Rhodes grass (Chloris gayana), kikuyu (*Pennisetum clandestinum*) and Bahia grass (*Paspalum notatum*).
e. However, *Desmodium uncinatum* can also become invasive,
f. As *Desmodium uncinatum* does not vigorously establish in the early stages of growth, weed competition should be prevented and

Fertiliser

It needs high soil fertility, and P, S and K must be applied before sowing and for maintenance. Silver leaf desmodium may recover from fire but fires are not recommended.

Yield

a. Silverleaf desmodium yields between 3 and 7 t DM/ha when it is grown in pure stands
b. Yields up to 15t DM/ha for silverleaf/grass mixtures.

Utilisation

a. *Desmodium uncinatum* is mainly used as fodder. It can be used for pasture, deferred feed, cut-and-carry and hay.
b. it should not be grazed before 16 weeks after sowing.
c. Heavy grazing is not suitable. Continuous grazing should be avoided
d. rotational grazing should include rest periods of between 4 and 8 weeks
e. It should not be cut below 10 cm since this can hinder re-growth and result in its destruction
f. It is also used for ground cover and mulch.

3.4 Lablab Beans (*Lablab purpureus*)



Description

a. is a summer-growing annual or occasionally short-lived perennial forage legume.
b. It is a twining, climbling, trailing or upright herbaceous plant that can be up to 3-6 m in length.
c. It has a deep taproot
d. Lablab is a summer growing legume that remains green during the dry season when other fodder is scarce and dry

Propagation

a. Lablab grows in places where daily temperatures are in the range of 18-35°C
b. annual rainfall is between 650 mm and 2500-3000 mm.
c. Lablab withstands high temperatures.
d. drought hardy and can grow during the dry periods of the year due to its taproot which can extract water from 2 m below the soil surface.
e. Lablab tolerates some flooding periods but does not withstand poor drainage or prolonged waterlogging.
f. Lablab thrives in a wide range of soils from poor sandy soils to heavy clays when drainage is good,
g. pH is between 4.5 and 7.5 and there is no salinity.
h. Lablab does better in full sunlight.

Establishment

a. Lablab is a fast growing legume that can provide fodder less than 3 months after sowing. It is suitable in intercropping systems where it is grown with maize, pearl millet and sorghum.
b. Lablab requires a well prepared seedbed
c. can be drilled at 3-10 cm depth or broadcast.
d. When it is intercropped, the seeds should be sown in well-spaced rows between maize or sorghum rows
e. it should be sown at the same time or when maize is already 15 cm high.
f. Lablab could increase the yield of intercropped maize even when the upper part of the plant was cut for livestock feeding.
g. Lablab outcompetes other legume forages such as velvet bean or cowpea in dry conditions.



Lablab seed

Utilisation

a. When used for fodder, lablab can be grazed
b. Is harvested for cut-and-carry systems,
c. Used For making hay and silage.
d. can be grown with other crops such as maize to make mixed fodder.
e. A N-fixing legume, lablab is a valuable green manure.
f. Lablab is used in ethnoveterinary medicine, for instance to treat eye problems in sheep and lung problems in sheep, cattle and goats.



Cattle grazing in a lablab field

3.5 Velvet Beans (Mucuna pruriens)





a. Velvet bean is a leguminous vine.
b. It is annual or sometimes short-lived perennial.
c. Velvet bean is vigorous, trailing or climbing, up to 6-18 m long.
d. its growth can suppresses companion species.

e. It has a taproot with numerous, 7-10 m long, lateral roots.
f. It is a fast growing
g. Velvet bean is a N-fixing legume that has no specific rhizobium requirements, but N fixation is favoured by warm temperatures.
h. Velvet bean requires a hot moist climate with annual rainfall ranging from 650 to 2500 mm
i. It can grow on a wide range of soils, from sands to clays but thrives on well-drained, light textured soils of appreciable acidity

Propagation

a. Does not require a high degree of land preparation.
b. Best results are with drilling
c. Plant seed 1 m between rows and 20–80 cm between plants (20–40 kg/ha seed)
d. Seeds are large and so seeding depth can be as deep as 10 cm but mostly 3–7 cm.
e. Seed does not require scarification or inoculation with rhizobia prior to planting.



Velvet beans planted in rows



Pods



Seed

Yield

Velvet bean yields range from 10 to 35 t green material/ha or 5.2-16.4 tonnes of DM/ha
It also yields from 250 to 3300 kg seeds/ha depending on the cultivation conditions

Utilisation

a. Velvet beans have three main uses: feed (forage and seeds) food, and environmental services.
b. Velvet bean is a valuable fodder and feed legume.

c. Vines and foliage can be used as pasture (when the pods are still young), hay or silage for ruminants
d. pods and seeds can be ground into a meal and fed to both ruminants and monogastrics.
e. Pods with their seeds can be ground into a rich protein meal and can be fed to all classes of livestock though in limited amounts in monogastrics.
f. Seed have anti-nutritional factors which can be reduced by boiling or soaking the seed.
g. improve soil fertility: it provides more than 10 t DM aboveground biomass/ha,
h. and below ground it fixes some 331 kg N/ha, equivalent to 1615 kg ammonium sulfate/ha
i. Velvet bean also yields 100 kg K/ha, and 20 kg P/ha
j. Can improve maize yield by between 500 kg and 2 t/ha after a one-year fallow.



A field of velvet beans

3.6 Jack Beans (*Canavalia ensiformis*)





Jack bean is a climbing perennial legume commonly cultivated as an annual.

Optimal annual rainfall is between 800-2000 mm but its deep rooting system allows it to withstand dry periods. It is reported to be tolerant to waterlogging and salinity

Propagation





a. Best results are with drilling
b. Plant seed 1 m between rows and 20–80 cm between plants (20–40 kg/ha seed)
c. Seeds are large and so seeding depth can be as deep as 10 cm but mostly 3–7 cm.

Utilisation

a. The whole plant, the pods and seeds are also used to feed animals.
b. Pods and seeds are edible and used for food, the young pods being cooked as a vegetable.
c. Jack bean is used in conservation agriculture with maize and cassava.
d. It is generally recognized as a soil improver. it is ploughed in for green manure for sugar cane after flowering.

3.7 Cow Peas *(Vigna unguiculata)*



a. The cowpea is an annual herbaceous legume cultivated for its edible seeds or for fodder.
b. It may be climbing and erect, as well as prostrate and creeping depending on the cultivar. Prostrate varieties grow to about 80 cm and climbing cultivars up to 2 m.
c. It has a well developed root system.
d. Seeds can be white, pink, brown or black.
e. Cowpea grows in savannah vegetation at temperatures ranging from 25°C to 35°C
f. and in areas where annual rainfall ranges from 750 mm to 1100 mm
g. Cowpea is tolerant of shading and can be combined with tall cereal plants such as sorghum and maize
h. Cowpea grows on a wide range of soils provided they are well drained
i. It is sensitive to waterlogging, though less than other legumes
j. High moisture may hinder cowpea crops in the sub-humid tropics due to the many diseases they are susceptible to.



Cowpea seeds



Cow peas planted in rows

Utilisation

a. Cowpeas are primarily valued as food,
b. Dual-purpose varieties have been developed to provide both grain and fodder
c. Cowpea forage, both the vines and leaves, are used either fresh, or conserved as hay or silage,
d. Cowpea pod husks obtained after threshing are also used to feed livestock.

4.0 Production of Tree Fodder Forages

4.1 *Gliricidia Sepium*





a. Small to medium-sized, thornless, leguminous tree up to 10-12 m high

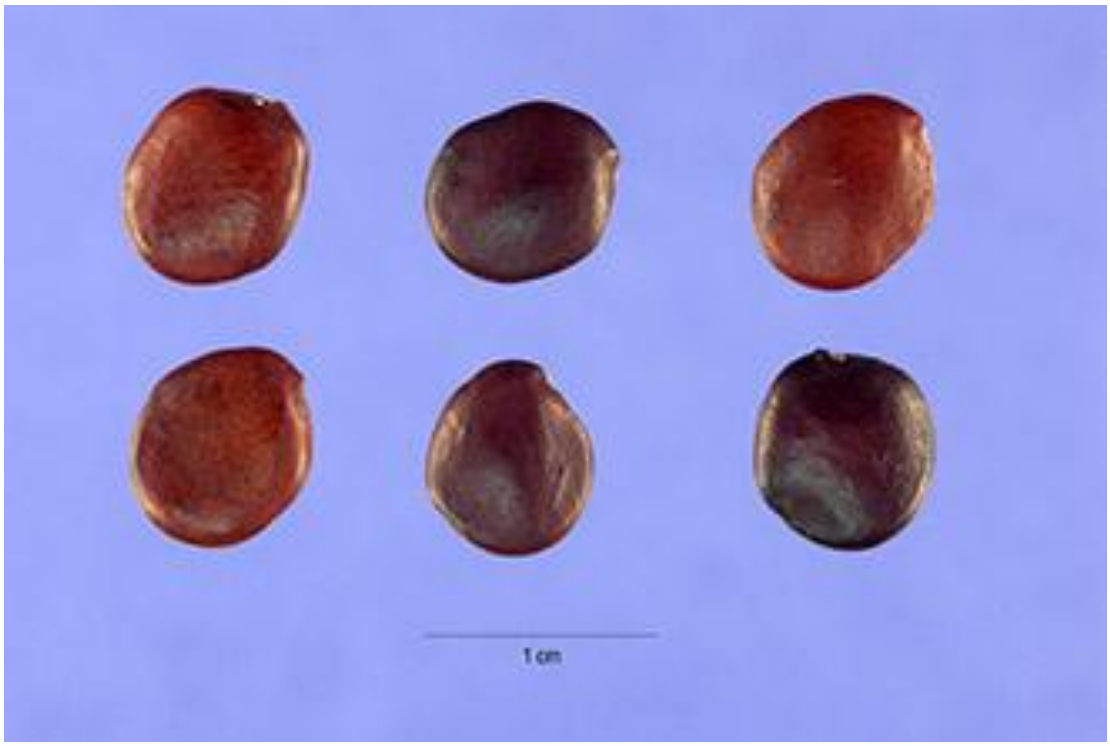
b. Is one of the major tropical forage trees due to its protein-rich forage and high nutritive value
c. is a perennial, medium-sized (2-15 m high) legume tree.
d. It is mostly deciduous during the dry season but is reported to remain evergreen in humid areas.

Propagation

a. It grows well from sea level to an altitude of 1600 m,
b. in areas where the mean temperature ranges from 20°C to 29°C,
c. and annual rainfall is between 900 mm and 1500 mm, with a five-month dry period.
d. It does not withstand frost and night temperatures below 15°C.

Soil Requirements

a. It is tolerant to waterlogging and to a wide range of poorly fertile soils.
b. Adapted to a wide range of well-drained soils.
c. often found on highly eroded soils of volcanic origin with pH 4.5-6.2,
d. but is also found on sands, heavy clays and slightly alkaline, calcareous limestone soils.
e. is suitable for acid, infertile soils.
f. Sowing depth for seeding into the field or nursery beds is 2 cm.
g. Scarification is unnecessary, and germination rates of >90% are typical.
h. Seedlings establish rapidly,
i. generally reaching a height of 3 m before flowering at 6-8 months of age. Trees can also be established rapidly from cuttings, using stakes of 5-6 months of age,
j. stakes should be 1.5 m long and with a diameter of 3.5-4.0 cm.
k. For densely planted protein banks, use stakes 50 cm long and six months of age.
l. Plant populations range from 4,000-10,000 trees/ha.
m. Very high densities are used in small protein banks. T
n. the direction of planting should be east to west to maximise sunlight interception.
o. Gliricidia does not grow well on wet or waterlogged soils



Gliricidia seeds



Gliricidia Seedlings

.

Yield

a. Yields 9 to 16 t/ha of DM in fodder plots,
b. It can be lopped around 7 months after establishment on plants grown from cuttings and 14 months after seedling.
c. Thereafter lopping can be done every 2 to 3 months during the rainy season and every 3 to 4 months during the dry season, provided regrowth reaches 1-2 m high before harvest.

Feeding

a. can be cut by hand and left on the ground for grazing or carried to paddocks or stalls.
b. High nutritive value. Crude protein content 18-30% and in vitro digestibility of 60-65%. With the exception of palatability , variability in nutritive quality among provenances has not been assessed.
c. It is also possible to make silage from chopped forage, which may be mixed with grasses or maize.
d. Additives, such as molasses, sugar cane or formic acid (0.85%), should be added to provide fermentable carbohydrates.
e. It may be used as the sole feed in the dry season. Feeding levels have been 1-3% of body weight for cattle and goats.



Feeding Sheep



Feeding Cattle

Other Uses

a. The plant is thus considered as a good soil improver as a legume it is able to fix N.
b. It produces a lot of litter and the half-life of gliricidia leaves is about 20 days.
c. it is used as a windbreak.
d. It thrives on steep slopes and may be used to reclaim denuded land.
e. is also often used as shade for perennials (coffee, tea, cocoa)or as nurse-tree since it produces light shade and reduces soil temperatures

4.2 Leucaena (*Leucaena Leucocephala*)

A thornlesss, productive, persistent leguminous shrub or small tree





a. High protein fodder. (28-31% CP for dried leaflets).
b. Carotene content can be twice as high as for lucerne.
c. Most of Zambia lies within the climatic limit required for it's growth at altitudes above 1200m growth will be reduced.
d. Extremely drought resistant and will retain green leaf throughout the dry season.
e. Leucaena is very palatable.
f. Leucaena is not affected by light frosts. Severe frost will cause loss of leaf but the plant is unlikely to be killed.
g. It can fix nitrogen.
h. Can be grazed up to 20 years.
i. Seed production is very prolific and hand harvesting is very simple.

Problems

a. Has mimosine (up to 12% DM in young shoots), which can make the animals lose hair and weight if they eat too much.
b. Initial growth and establishment is slow and may take a whole season and even several years to establish.

Establishment

a. can grow on a wide range of soils
b. Established plants will tolerate water – logged conditions
c. Leucaena should be grown in spaced stands..

Needs Scarification

a. Place in very hot water (80^0 for 2 minutes) and then drying.
b. Do not boil the water.
c. When only a small number of seeds are to be planted, the broad end of the seed can be rubbed on sand paper or rough stone to break the seed coat.

Needs inoculation

a. It is essential that seed be inoculated with the correct rhizobial inoculum, before sowing, otherwise growth will be poor.
b. Alternatively soil from beneath leuaena trees that are known to be nodulating effectively can be used. This method, however, is not so reliable.

Sowing

a. Sow seed at a depth of 2cm as soon as the rains have become established.
b. Emergence will take 10-14 days.





Leucaena seeds

Can also be established by transplanting 1-2month old seedlings or when the plants are about 15cm high



Leucaena seedlings

Spacing requirement of Leucaena

Utilisation	Seed Rate Kg/ha	Spacing Between rows (M)	Spacing; Within Rows (cm)
Forage Harvester	30	0.5-0.75	10
Hand Cutting	17-30	1.0	15-30
Grazing	11-22	3.0	10

Fertiliser

a. Basal application 400kg per ha single supers at planting.
b. Annual top dressing 200kg per ha D Compound.

Leucaena Weed and Pest Control

a. Leucaena seedlings are very susceptible to weed competition. They should be kept free of weeds for the first three months after emergence.
b. Termites can be a problem to young plants. These should be controlled

Yield

a. Average yield ranges from 3 to 30 t DM/ha/year depending on soil, temperature and moisture conditions.
b. For optimal yields, harvest interval can vary from 6-8 weeks in very productive sites to 12 weeks in less productive ones.

Utilisation

a. Cut and feed to animals
b. Take the first cut when flowering starts and subsequent harvests when the branches are about 1m in length.
c. Cut re-growth back to a height of 15-30 cm.
d. Grazing. Leucaena may be lightly grazed in the first year after seedling and heavily grazed after the second year. However grazing time should be limited to animals to avoid poisoning (mimosine)
e. Poisoning occurs if it is more than 30% of the diet and fed over long periods
f. Reduce poisoning by supplementation with zinc sulphate or Fe salts alleviates leucaena toxicity. Mimosine content can also be reduced by soaking in water and drying.
g. Can also be dried, milled and mixed with other feedstuffs



Livestock browsing on Leucaena

Leucaena's contribution to the environment are manifold:

a. N-fixing: it fixes large amounts of N (150 to 300 kg/ha) thus promoting grass or maize growth.
b. Erosion control and land reclamation. Its deep taproot helps breaking up compacted subsoil layers thus improving water penetration and reducing surface run-off.
c. It prevents saline subsoil water from reaching the surface. Grown in contour strips, leucaena helps in erosion control on steep slopes.
d. It provides shade and acts as a shelterbelt (preventing wind damages) to other trees such as cocoa, coffee and tea but also to climbing crops (vanilla, pepper, passion fruit).
e. It provides green manure in alley cropping systems since its leaves decompose quickly.

4.3 Pegion Peas (*Cajanus cajan*)





a. An erect woody, annual or short-lived perennial shrub or small tree, 1–4 m tall with a deep taproot (to 2 m).
b. One of the most common legumes of the tropics and subtropics because of its very wide adaptability.
c. Deep rooted and drought resistant, growing especially well on semidry land.
d. Because of its poisonous roots, often planted as a hedge around cassava patches to keep out mole rats, and sometimes around houses for protection against termites.

Soil requirements

a. Can be grown on a wide range of soil textures, from sands to heavy black clays
b. needs free-drainage.
c. prefers pH of 5–7, but can tolerate pH 4.5–8.4. Can handle some Al.
d. Pigeon pea is sensitive to high salinity.
e. As a crop, pigeon pea is very drought tolerant, able to grow with a dry season exceeding 6 months, and rainfall <300 mm, but does best with 600–1,000 mm rainfall , and where elevated, in excess of 2,000 mm.
f. Less suited to the humid tropics with poor tolerance of wet soils or flooding.

Propagation

a. Can be grown as an intercrop with cereals or legumes, which are harvested to avoid competition before the pigeon pea flowers and sets seeds.
b. Pigeon pea does best when broadcast and covered or drilled into a well-prepared seedbed in rows 35 cm apart, at a depth of 2.5–10 cm
c. seeding rate of 0.5–4 kg/ha, or 5–7 kg/ha under irrigation.
d. Can be planted into holes on a 2 m grid as a crop. Broadcast or planted in rows with a seed rate of over 20 kg/ha when sown as ground cover or for fodder.



Dry seed

Fertilizer

Although a nitrogen fixing species, fertiliser is recommended at levels of 20–25 kg/ha N. Phosphorus is the most limiting factor for pigeon pea and applications of 17–26 kg/ha P can increase seed yield by 300-600 kg/ha. Most of the pigeon pea cultivars are susceptible to zinc deficiency. Applications of 2–4 ppm zinc as 0.5% zinc sulphate with 0.25% lime have been effective to over-come zinc deficiencies.



A clean field of pegion peas

Compatibility (with other species)

a. As a forage shrub , pigeon pea can be interplanted with grasses.
b. Tall late flowering varieties are commonly grown as a food grain crop intercropped with sorghum or millet in the semi-arid tropics.
c. Has grown well with *Chlorisgayana* and been planted into pangola grass (*Digitariaeriantha*) pastures.

d. When used in intercropping systems, the deep root system competes less with associated crops than some other legumes

Dry matter

a. In semi-arid conditions in the drylands of Africa, fodder yield is usually in the order of 2 t/ha.
b. However, DM yields may be up to 25 t/ha and may reach up to 40 t/ha with 100 kg/ha N under optimal conditions.
c. Leaves remain on the plant under dry conditions.
d. Good yields of wood for fuel although calorific value is low.

Nutritive value

a. High quality leaf (10–15% crude protein), especially when developing seedpods are mixed with the edible stem fraction.
b. Cattle on pigeon pea pasture gained 0.7–1.25 kg/day at carrying capacity of 1–3.75 cattle/ha, with liveweight gains of 200–500 kg/ha/yr .

4.4 Moringa (Moringa oleifera)





Tiny leaves.

Enormous benefits.



=

7 times the Vitamin C of Oranges

4 times the Vitamin A of Carrots

4 times the Calcium of Milk

3 times the Potassium of Bananas

2 times the Protein of Yogurt



100g Moringa Dry Leaf =

10 times the Vitamin A of Carrots

½ times the Vitamin C of Oranges

17 times the Calcium of Milk

15 times the Potassium of Bananas

25 times the Iron of Spinach

9 times the Protein of Yoghurt

These figures reflect gram-for-gram comparisons with Moringa leaves.

a. Moringa is a small to medium evergreen or deciduous tree that can reach up to 10-12 m.
b. is a multipurpose tropical tree.
c. It is mainly used for food and has numerous industrial, medicinal and agricultural uses, including animal feeding.
d. Nutritious, fast-growing and drought-tolerant,
e. its cultivation has since become increasingly popular in Asia and Africa,
f. it is among the most economically valuable crops. It has been dubbed the "miracle tree" or "tree of life" in popular media

propagation

a. Moringa is propagated from seeds or from cuttings.
b. When it is grown from seeds it can be sown either directly or in containers.
c. Moringa should be sown at the beginning of the rainy season on elevated seedbed so that it is protected from waterlogging but it can benefit from soil moisture.
d. propagation through seeds
e. propagation from cuttings is suitable for high-grade fruit production and for root production
f. The recommended spacing between moringa plants depends on their final use and geographic location. Leaf production increases with sowing or planting densities
g. Spacings as small as 5 x 10 cm to 20 x 20 cm were successfully used for leaf production in home-gardening conditions
h. Larger spacing (75 cm x 100 cm) is recommended when moringa is sown on larger areas so that the number of seeds to buy is not too important
i. However, it was reported that intensive production on large fields was possible and yielded up to 44 t/ha of DM
j. When moringa is cultivated for pod production, 2.5 x 2.5 m spacing is recommended This level of spacing is also suitable when moringa is intercropped with another plant species
k. Moringa can be grown in pure stands or in mixture with many other tree species such as leucaena (Leucaenaleucocephala) and with vegetables species in alley cropping, or with grasses or cereal crops



A Plantation of Moringa

Harvest and yields

a. Moringa has an outstanding growth and can be harvested for foliage in less than 2.5 months.
b. Optimal cutting intervals range from 15 to 75 days, depending on local conditions
c. Fodder yields are very variable and may range from 27 to 120 t/ha of fresh matter at first cutting (
d. Up to 9 cuttings/year could be done
e. Highest growth and forage yield are obtained under warm, dry conditions, with some supplemental fertilizer and irrigation
f. Plant management is important for leaf production. Pollarding and coppicing promote leafy regrowth and enhance leaf yield
g. Moringa trees grown for pod production can be harvested 7 months after planting.
h. Pod yields are about 19 kg pods/tree/year, somewhat equivalent to 30 t pods/ha/year.



Increases daily weight gain up to **32%**

Increases milk production **43%** to **65%**

FORAGE CONSERVATION AND UTILISATION

5.0 Forage Conservation and Utilization

Why conserve forages?

During the rainy season (Nov-March), there is plenty of forages for grazing animals. This makes them healthy, gain weight, reproduce, and produce a lot of milk for their young ones and also for family consumption.



Rain Season



Dry Season

However, when dry season starts (May –Sept), grass dries up and some get bunt by bush fires, making feed for these animals scare or if it is available, of poor quality.



Animals start losing weight in the dry season due to less feed available

As a result, animals lose weight, some become sick and even die, most of them fail to reproduce,

Also, weak or sick animals are unable to produce enough milk or meat, and cannot be used as oxen for ploughing during cultivation of crops, or transporting things in an oxcart.

When selling, they are sold at very low prices, which makes the farmer loose income.

Therefore, there is need to conserve forages during time when they become abundant in rainy season to feed the animals when these forages are dry or scarce in the dry season.

So, what can you do to counter feed shortages during dry seasons?

You have several options and opportunities to counter the feed shortages that annually occur during dry seasons. Some of the options include:



- ✓ Planting fodder crops for supplementary feeding during the dry season (e.g. sunhemp, velvet beans, cow peas, mulberry trees, moringa, etc)

Planting improved pastures for grazing (e.g, Rhodes grass, sudan grass, napier grass).This, however, is only suitable for farmers on non-traditional land.



Cutting natural grass or improved pasture and transporting it for direct feeding to the animals if they are few in number.



Chopping fodder for feeding animals or for making silage



- ✓ Or intercropping legumes in pasture/grass stands to improve nutritional quality of the grass.



5.1 How to conserve forage

How should forages be conserved for the dry season?

Forages can be conserved to feed livestock during periods of shortages in quantity and quality. Forages, whether pastures, fodder crops or crop residues can be conserved in two forms:

(i) Silage (ii) Hay

What is Silage?



This is the product of cut forage of high moisture content which has been fermented to produce a stable feed which resists further breakdown in anaerobic storage

What is Hay?



Hay is plant or vegetative material preserved by drying for future feeding to animals. The idea is to quickly reduce moisture content from 80% to 10-15%.

At this moisture level the grass does not rot in storage as all biological activities are arrested.

Good quality hay is obtained from what is cut before flowering in order to preserve the much needed nutrients which can rapidly decline if the crop "goes to seed" or ages.

However, in whatever form forage is conserved, it is important to remember that, at best, they can rarely match the nutritive value of fresh forage.



You can also make hay from natural pasture (veldt), provided they are cut early enough (by March) to conserve the nutrients.

However even in March, it is often too wet to dry the pasture successfully and special machinery, which is very expensive, has to be used to assist the forage to dry quickly.

Other forage crops such as maize, and forage sorghum are too thick-stemmed to dry successfully as hay. It is also difficult to make hay successfully from legumes as they drop their leaves very readily, either when handing the dried material or if allowed to get too mature before cutting. Therefore, silage is considered the better way to conserve these forage crops.

5.3 Silage

How is Silage made?

Silage making is the process of preserving forages for future use while maintaining the nutrient contents. The objective is to retain the nutrients present in the green forage and deliver a silage accepted by livestock. This is usually attained through an anaerobic fermentation dominated by lactic acid bacteria. This bacteria (present on the crop and in the air) produce acid, which reduces the pH of the grass to approximately 4.2. This prevents decomposing and deterioration of fresh green grass and preserving its quality.

Many forage crops can be ensiled including ones shown below:



Legumes such as velvet beans



Maize



Sorghum



Napier grass



Naturally growing grass



Rhodes grass

However, maize silage is preferred to the rest due to the high tonnage produced per acre and also the high nutritional content when harvested at the correct time.

Steps to follow when making silage:

Step 1: **Harvesting**



- Harvest the material (grass, fodder crops or legumes) to be used from the field.

Cutting grass using a slasher

There are so many tools that you can use to harvest the grass or forages for silage making:



You can use a sickle, scythe, or simply a slasher

A rule of thumb for harvesting forage is, the more mature a crop is, the lower the nutrients it possesses. This is because as the crop matures, it has less leafy content and more "harder" parts such as a stronger stem resulting in lower nutrients and hence it is important that a farmer harvests the crop at the right time.

After cutting the materials in the field, transport them to the site where you will do the silage making. To transport it, you can use a bicycle



- An oxcart, or tractor

Step 2: ***Wilting***

Wilting is the process of making the crops loose some moisture.





- This can be done by spreading the cut plants on the ground or lay them against the wall or on a rack outdoors and in the sun.
- When wilting, it is important to remember not to wilt in thick layers as the plants at the bottom will not be exposed to the sun/heat and may lead to decomposition undoing the silage making process. Turning severally is also advised.

The crops should be wilted to around 30% dry matter. A simple test to determine whether the forage is wilted enough; squeeze the chopped material in your hand. If there is moisture, the /hand is wet but no water dripping and it does not return to its original form quickly, then the crop is well wilted ready for silage making. Some farmers smash the nodes and stems to fasten the wilting rate.

Step 3: **Chopping**



- Chop the wilted plants to lengths of between 1-3cm. This is because shorter chop lengths allow for better silage making.

Also, longer chop lengths are more difficult to compress and displace the air within the crop especially the stems which are harder and hence more difficult to compress.



- Chop forage using a panga;
- Take caution that you do not injure yourself



- A simple locally made machine.
- Also take caustion that you do not cut yourself when slicing the forage.



- Or using a manually operated machine

Step 4: **Compacting chopped forage**

This stage may be done in two ways.



- You may want to mix molasses with the chopped forages as you fill and compact it in the container.
- Or you may prefer to compact it as it is without adding molasses. This is common when ensiling maize when you opt to utilize the natural sugars in the maize plant.

Whatever you decide, compacting must be done well, thoroughly and layer by layer.



- What you are trying to achieve is to expel as much air as possible and to maintain the condition as is without oxygen as possible. Compact the chopped forages as you fill the plastic bag.



- The containers can be simple shopping plastic bags. When you have finished filling and compacting, squeeze the air out after filling the first bag layer and tie tightly.



- To strengthen the plastic bag, put a second layer of bag and tie it tightly



- Put a third layer bag and tie it tightly as well
- These three layers of plastic bags are adequate not easily tear and allow air in.

To avoid stretching and tearing of plastic bags when compacting, use a removable mould (e.g. half plastic drums). This makes compaction easy and protects plastic from stretching and tearing.





You can also use large containers, for example full drums or a silage pits.

When using drums, you have to first line it with plastic bag inside.

Then, start filling the materials, compacting layer after layer



- You can compact by stepping in as you fill the chopped forage



- After compacting, tie the mouth of the bag and remove from the mould.
- Store the bags safely until feeding time sometime later

You can also ensile forages in pits dug out on the ground.



- First line it with a plastic cover the bottom and sides of the pit
- You can fill in with the chopped materials, and after a certain layer, compact it.



- This can be done by rolling a drum filled with water over the chopped forages in the pit.

Once the pit is filled up, cover it with the overlapping plastic and put some heavy objects on top to ensure that no air enter your silage material in the pit.

The silage will ready after 60 days, and if properly done, it can be stored for many years.

5.4 Hay

Hay is forage dried by sun and wind, which is conserved as standing hay or cut. Good quality hay is obtained from what is cut before flowering in order to preserve the much needed nutrients which can rapidly decline if the crop "goes to seed" or ages. Depending on fodder crop from which the hay is made, the following are the different types of hay:



Legume Hay;

- This has high feeding value and is especially rich in digestible crude protein, vitamins and minerals.
- It is highly palatable. Examples are cowpeas, soya beans, velvet beans, and sunn hemp hay.



Non-Legume Hay:

- This is hay from predominantly grasses and it is generally less palatable than legume hay.

Generally, the quality largely depends

on the stage of harvesting. Protein content, minerals and vitamins in grasses are usually lower than in legumes and examples are Rhodes grass, buffel grass, natural grass, and maize stover.

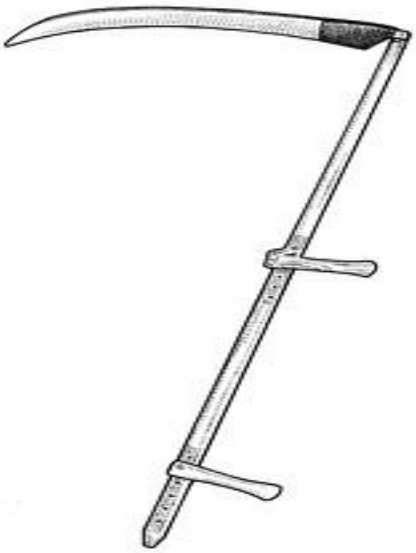


Mixed Hay:

- This is hay made from mixture of grass and legumes.
- The nutritive value depends on the type of mixture and the ratio of legumes and non-legumes used in the mix.

Examples of mixtures are Cowpeas and Rhodes grass or maize, Velvet beans with sorghum and/or maize and/or Rhodes grass.

To make good quality hay, Harvest at the end of the rainy season, in March or April when there is plenty of sunshine and when the grass fodder is still green and tender.

	You can cut the fodder with a sickle though causes some back pains because bending when doing the job
	Or specially made fodder cutter called a Scythe. This tool is especially suitable for women as it eliminates bending (as in the case of the traditional sickle).

Whichever tool you use, the best stage to cut the forage is when half of the fodder has started flowering. This will optimize both quality and quantity of grasses and legume.



- After cutting, dry the fodder as quickly as possible. Spread the fodder in the field to dry for about 4 hours.



- Once the top is dry, use a rake to turn the fodder and dry again for approximately 4–5 hours.

Follow the same process the following day, provided there has not been rain. The hay can be removed from the field for storage the second day. As a guide, hay is ready to store when no sap (moisture) can be drawn from the stem when pressed with the fingernail.

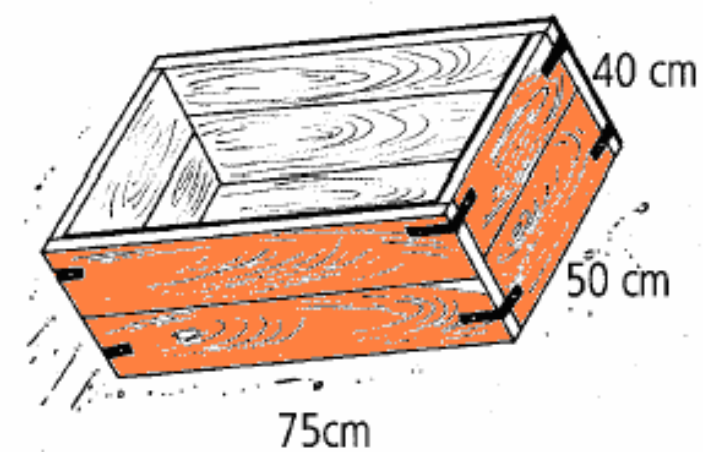
Handle the dried hay, for example natural grass, gently, as the leaves especially of legumes become crispy and easily break from the plant, which leads to loss of the most nutritious part of the plant.

To prevent loss of feeding value, the drying period should be as short as possible and after drying, hay should be handled as little as possible.

Once dried, hay can be gathered loose or can be baled for easier storage and transport. Baling is strongly encouraged as it saves storage space, avoids wastage at feeding and facilitates easy measurement for sale/marketing.

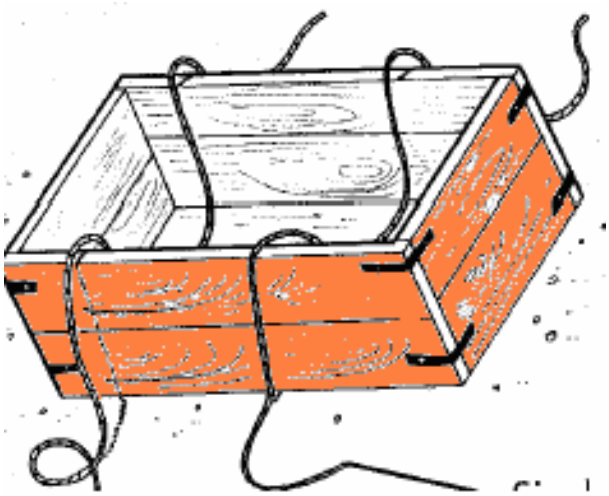
To bale hay the following are the steps to follow:

Step 1:



- Make a box measuring, for example 50cm width, 75cm length and 40cm in depth, as shown in the picture.

Step 2:



- Put strings in the box as shown in the picture. The strings must be strong enough to hold compacted grass together.

Step 3:



- Fill in the dried grass and press the hay tightly in the box by stomping feet on it or using any heavy object, such as a metal bar, to compact the hay.

Step 4:



- Tie the bale with the string tightly.

Step 5:



- Lift the box to release the bale.



Step 1: *Put strings in the box*



Step 2: *Fill in the dried hay and compact it*



Step 3: *Tie the strings tightly*



Step 4: *Lift the box to release the bail*

Where a wooden box is not available, a pit with the same measurements as the size of the bale desired can be made in the group and used as a mould.



- Dig a pit on the ground measuring, for example 100 x 60 x 60cm (Length, width and height).



Put strings in the pit for tying the hay as shown.



- Fill in as much grass as possible while compacting.



- Compact the grass by trampling on with your feet.



Tie the compressed grass in the pit tightly



Bale of hay removed from the pit.

Storage of hay



- The bales should be stored under a roof and stacked on raised wooden structures. The hay rack provides sufficient ventilation and it helps avoid spoilage caused by rain, sun, or insects like termites.







6.0 Forage utilization

Silage removed from a pit or emptied out from a bag should be fed as soon as possible, preferably within a few hours. After this it can be mouldy and become unpalatable because of toxins formed in the silage when it is exposed to air. After feeding, the feed bunks must be cleaned out to prevent any remaining silage, which will spoil, contaminating the next feed out.



Feeding Silage

Hay Feeding

In order to improve the taste and intake and nutritive value of the hay salt and Molasses can be added. 100grammes of salt per 10kg of hay should be used. The salt should be first be dissolved in water and the water sprinkled on the hay.





